

- 6.9 The practical Semester End Examination shall be conducted by two examiners, one of them being internal examiner and the other being external examiner. The external examiner shall be appointed by the Principal with recommendation by the Head of Department.
- 6.10 For the course having design and/or drawing, (such as Engineering Graphics, Engineering Drawing, Machine Drawing) and Estimation, the distribution shall be 30 marks for internal evaluation (20 marks for day-to-day work & 10 marks for mid examinations) and 70 marks for Semester End Examination. There shall be two internal tests in a Semester. Student's performance in both the MID examinations will be considered for evaluating the internal marks. The best scored MID exam weightage will be given as 70% and for the other 30% weightage will be given.
- 6.11 There shall be a Minor-Project in II Year II Semester. The project work could be done in the form of a minor practical project in the college. Participation in any technical event/competition to fabricate and demonstrate an innovative machine or product could be encouraged under this course. The minor project shall be submitted in a report form and presented before the committee. It shall be evaluated for 100 marks. The committee consists of Head of Department, the Supervisor of the Minor-project and a Senior Faculty member of the department. There shall be no external examination for minor-project.
- 6.12 There shall be Technical Term Paper in III Year II Semester. A technical term paper can be a research paper/report required at the end of a semester.
- a) A student at the end of the semester will submit technical report / document that describe the process, progress, or results of technical or scientific research or the state of a technical or scientific research problem. It might also include recommendations and conclusions of the research.
 - b) Technical report will be evaluated for 100 marks in total. Assessment will be done for 100 marks out of which, the supervisor / guide will evaluate for 30 marks based on the work and presentation / execution of the work. Subdivision for the remaining 70 marks is based on publication/report/presentation/execution, etc which will be decided by the Project Review Committee.
 - c) Evaluation shall be done by a committee comprising the Head of Department, the Supervisor and a Senior Faculty member of the department. There shall be no external examination for Technical Term paper
- 6.13 There shall be a Mini-Project, in collaboration with an industry of their specialization, to be taken up during the vacation after III Year II Semester examination. It shall be evaluated along with the project work in IV Year I Semester. The Mini-Project shall be submitted in a report form and presented before the committee. It shall be evaluated for 100 marks, of which there shall be continuous evaluation during the semester for 30 internal marks and 70 Semester End Examination marks. The committee consists of an External examiner, Head of the Department, the Supervisor of the Mini-project and a Senior Faculty member of the department.
- 6.14 There shall be Major-Project Phase-I, in IV Year I Semester. The objective of Project Work Phase-I is to enable the students to take up investigative study in the broad field of respective branch of Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the department to two/three/four students in a group, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work. The team shall submit the work in a report form and present the same before the committee. It shall be evaluated for 100 marks, of which there shall be internal evaluation during the semester for 30 internal marks and 70 Semester End Examination marks. The committee consists of an External examiner, Head of Department, Coordinator of Major-project, project guide and a Senior Faculty member of the department.
- 6.15 There shall be Major-Project Phase-II, in IV Year II Semester. The objective of Project Work Phase-II is to enable the students to extend further the investigative study taken up under Project Work Phase I, either fully theoretical/practical or involving both theoretical and practical work, under the guidance of a Supervisor from the department alone or jointly with a

B V RAJU INSTITUTE OF TECHNOLOGY

II Year B.Tech. I.Sem

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DIGITAL LOGIC DESIGN

PRE-REQUISITES: Electronic Devices and circuits

COURSE OBJECTIVES:

This course provides in-depth knowledge of switching theory and the design techniques of digital circuits, which is the basis for design of any digital circuit.

1. To learn basic techniques for the design of digital circuits and fundamental concepts used in the design of digital systems.
2. To impart to student the concepts of sequential circuits, enabling them to analyze sequential systems in terms of state machines.
3. To Design Complete Digital System.

UNIT-I

Number System

Number Systems, Base Conversion Methods, Complements of Numbers, Signed Numbers-Signed Magnitude, 1's Complement and 2's complement representation, Signed and unsigned addition /subtraction, Binary Codes-Binary Coded Decimal, Excess-3 code, Gray code and Error Detection Codes-Parity code, Hamming code.

UNIT-II

Boolean Algebra

Basic Theorems and Postulates, Digital Logic Gates, Universal Gates, Algebraic Simplification of Digital Logic Gates using Theorems and Postulates, Multilevel NAND/NOR realizations, Canonical and Standard Forms. The Karnaugh Map Method, Five Variable Maps, Prime and Essential Implications, Don't Care Map Entries, Partially Specified Expressions

UNIT-III

Combinational Design

Arithmetic Circuits-Half adder, Full adder, Half subtractor, Full Subtractor, 4-bit parallel adder/subtractor, BCD Adder, 2-bit comparator, 4:1 Multiplexers, Implement Boolean functions using Multiplexers, Decoders, Implement Boolean functions using Decoder, BCD to Seven Segment Decoder, Encoder-2 to 4 Encoder, Priority Encoder, Code Converters-Binary to Gray, Gray to Binary, Binary to BCD, BCD to Binary, Barrel shifter and ALU.

UNIT-IV

Sequential Circuits

Basic Architectural Distinctions between Combinational and Sequential circuits, The basic Latch-NAND latch & NOR latch, Clocked Latches-SR, D, JK and Flip Flops-Master-Slave F/F, Edge Triggered F/F, Flip-Flop, Excitation Tables, Conversion from one type of Flip-Flop to another,

UNIT-V

Design and Analysis of Sequential Circuit

Overview of Synchronous and Asynchronous Sequential Circuits, Finite State Machines –Mealy Machine and Moore Machine, State Diagram, State Minimization and Assignments, Design Procedure and Realization using Flip-Flops, Analysis of Synchronous Finite State Machines, Design of Asynchronous Counter-Ripple Up/Down Counter and Mod-N Counters, Design of Synchronous counters , Binary and BCD Up/Down counters and Mod-N Counters, Shift Registers-SISO, SIPO, PISO, PIPO, Bidirectional and Universal Shift Register, Shift Register based Counters-Ring and Johnson (Twisted Ring) Counter.

UNIT-VI

Algorithmic State Machines (ASM), Logic Families & Programmable Logic Devices (PLD)

Algorithmic State Machines-Components of ASM Charts, Salient features of the ASM chart and Design Examples using ASM Charts-Sequence Detector, Serial Adder. Logic Families-RTL, DTL, TTL, ECL & CMOS logic Families and their specifications. Programmable Array Logic (PAL), Programmable Logic Array (PLA), Combinational Circuit Implementation Using PAL and PLA.

COURSE OUTCOMES:

At the end of the course students will be able to:

1. Understand different number systems, codes and conversions.
2. Understand and optimize Boolean functions using K-Maps
3. Design combinational logic circuits
4. Understand basic building blocks of Sequential Logic circuits
5. Design & Analyze Sequential Logic circuits
6. Design Digital Systems using Programmable logic Devices and understand Logic Families

TEXT BOOKS:

1. Digital Design- Morris Mano, PHI, 3rd Edition.
2. Switching Theory and Logic Design- A. Anand Kumar, PHI, 5th Edition.

REFERENCES:

1. Switching and Finite Automata Theory- ZviKohavi& Niraj K. Jha, 3rd Edition, Cambridge
2. Introduction to Switching Theory and Logic Design --Fredriac J. Hill, Gerald R. Peterson, 3rdEd,John Wiley & Sons Inc.
3. Digital Fundamentals – A Systems Approach – Thomas L. Floyd, Pearson, 2013.
4. Digital Logic Design - Ye Brian and Holds Worth, Elsevier
5. Fundamentals of Logic Design- Charles H. Roth, Cengage Learning, 5th, Edition, 2004.
6. Digital Logic Applications and Design- John M. Yarbrough, Thomson Publications, 2006.
7. Digital Logic and State Machine Design – Comer, 3rd, Oxford, 2013.

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ELECTRONIC DEVICES AND CIRCUITS LAB

Course Objective

The objective of this course is to introduce the students about to provide an overview of the principles, operation and application of the basic electronic components. And Understand the Characteristics of the active devices and frequency response of different amplifiers.

PART A :Electronic Workshop Practice

1. Identification, Specifications, Testing of R, L, C Components (Colour Codes), Potentiometers, Switches (SPDT, DPDT, and DIP), Coils, Gang Condensers, Relays, Bread Boards.
2. Identification, Specifications and Testing of Active Devices, Diodes, BJTs, JFETs, Power Transistors, LEDs, LCDs, Optoelectronic Devices, UJT.
3. Soldering practice – Simple Circuits using active and passive components.
4. Single layer and Multi-layer PCBs (Identification and Utility).
5. Study and operation of Ammeters, Voltmeters, Transformer, Analog and Digital Multimeters, Function Generator, Regulated Power Supplies and CRO.

PART B: Experiments (minimum of 10)

1. PN Junction diode characteristics
2. Zener diode Reverse Bias characteristics and Zener Diode as a regulator
3. Half wave Rectifier (with & without filters)
4. Center Tap Full wave Rectifier with filters (with & without filters)
5. Bridge Rectifier with filters (with & without filters)
6. Transistor CB characteristics (Input and Output) & h Parameter calculations
7. Transistor CE characteristics (Input and Output) & h Parameter calculations
8. FET characteristics (Drain, Transfer characteristics) and calculate Drain Resistance (r_d), Trans Conductance (g_m), Amplification factor (μ).
9. Emitter Characteristics of UJT
10. Design and verify Self Bias Circuit. (Q - Point)
11. Frequency response of CE Amplifier (With and without Emitter bypass capacitor) and calculate Bandwidth, input and output impedances.
12. Frequency response of CC Amplifier (Emitter Follower) and calculate Bandwidth, input and output impedances.

COURSE OUTCOMES

After completion of the course the student will be able to

1. Identify and test different Passive Components & Active devices.
2. Understand the characteristics of the PN junction diode and Zener diode
3. Understand the operation of rectifiers with and without filters.
4. Obtain the input and output characteristics of BJT,FET,UJT and SCR.
5. Obtain the frequency response of BJT and FET Amplifier.

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DIGITAL DESIGN USING HDL

PRE-REQUISITES: Digital Logic Design

COURSE OBJECTIVES:

This course provides in-depth knowledge on

1. Fundamental building blocks of digital design across several levels of abstraction,
2. From CMOS gates to hardware design languages.
3. Important functions such as gates, decoders, multiplexers, flip-flops, registers, and counters are discussed at each level.

UNIT-I INTRODUCTION TO VERILOG

Verilog as HDL, Levels of Design Description, Concurrency, Simulation and Synthesis, Functional Verification, System Tasks, Programming Language Interface (PLI), Module, Simulation and Synthesis Tools, Keywords, Identifiers, White Space Characters, Comments, Numbers, Strings, Logic Values, Strengths, Data Types, Scalars and Vectors, Parameters, Memory, Operators

UNIT-II MODELLING AT GATE & DATA FLOW LEVELS

GATE LEVEL MODELING : Introduction, AND Gate Primitive, Module Structure, Other Gate Primitives, Illustrative Examples, Tri-State Gates, Array of Instances of Primitives, Additional Examples, Design of Flip-flops with Gate Primitives, Delays, Strengths and Contention Resolution, Net Types, Design of Basic Circuits.

MODELING AT DATA FLOW LEVEL : Introduction, Continuous Assignment Structures, Delays and Continuous Assignments, Assignment to Vectors, Operators.

UNIT-III BEHAVIORAL MODELING

Introduction, Operations and Assignments, Functional Bifurcation, *Initial* Construct, *Always* Construct, Examples, Assignments with Delays, *Wait* construct, Multiple *Always* Blocks, Designs at Behavioral Level, Blocking and Non blocking Assignments, The *case* statement, Simulation Flow. *if* and *if-else* constructs, *assign-deassign* construct, *repeat* construct, *for* loop, the *disable* construct, *while* loop, *forever* loop, parallel blocks, *force-release* construct, Event.

UNIT-IV SWITCH LEVEL MODELING & SYSTEM TASKS, FUNCTIONS, AND COMPILER DIRECTIVES

Introduction, Basic Transistor Switches, CMOS Switch, Bi-directional Gates, Time Delays with Switch Primitives, Instantiations with Strengths and Delays, Strength Contention with Trireg Nets, Parameters, Path Delays, Module Parameters, System Tasks and Functions, File-Based Tasks and Functions, Compiler Directives, Hierarchical Access, , User- Defined Primitives (UDP)

UNIT-V SEQUENTIAL CIRCUIT WITH HDL

Sequential models –feedback model, capacitive model, implicit model, basic memory components, functional register, state machine coding, sequential synthesis. Case studies: Design of FIFO, Counters, Shift registers

UNIT-VI COMPONENT TEST AND VERIFICATION

Test bench with Case studies -combinational circuit testing, sequential circuit testing, test bench techniques, design verification, assertion verification.

COURSE OUTCOMES

By the end of this course, students should be able to:

1. Describe Digital circuits at gate and Data flow level.
2. Describe Digital circuits at Behavioral level
3. Develop applications using Combination MSI Circuits for their academic projects
4. Develop applications using Sequential MSI Circuits for their academic projects

TEXT BOOKS:

1. T.R. Padmanabhan and B. Bala Tripura Sundari, Design through Verilog HDL , WSE, 2009.
2. Zainalabdien Navabi, Verilog Digital System Design, TMH, 2nd edition.

REFERENCES:

1. Fundamentals of Digital Logic with Verilog design by Stephen. Brown and Zvonko Vranesic, TMH, 2nd edition 2010.
2. Digital Logic Design using Verilog, State machine & synthesis for FPGA, SungguLee, Cengage Learning, 2009.
3. Verilog HDL- Samir Palnitkar, 2nd Edition,
4. Advanced Digital Design with Verilog HDL–Michael D. Ciletti, PHI, 2005.
Digital systems Design using VHDL – Charles H Roth, Jr. Thomson Publications, 2004.

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	HDL LAB	0	0	3 1.5

PRE-REQUISITES:Digital Logic Design.

COURSE OBJECTIVES:

1. Design & Practice the linear and non-linear applications of an op-amp & special application IC's
2. Operationalize different designs based on various IC's
3. Simulate digital systems design components using modern engineering tool.

Programming can be done using any compiler. Download the programs on FPGA / CPLD boards and performance testing may be done using pattern generator / logic analyser apart from verification by simulation using Equivalent front end CAD tools.

1. Case study of FPGA architectures
2. HDL code to realize all the logic gates
3. Design of 2-to-4 decoder
4. Design of 8-to-3 encoder (without and with Priority)
5. Design of 8-to-1 multiplexer and 1 x 8 Demultiplexers
6. Design of 4 bit binary to gray code converter
7. Design of 4 bit comparator
8. Design of Full adder using 3 modelling styles
9. Design of flip flops: SR, JK, T
10. Design of 4-bit binary, BCD counters (synchronous/ asynchronous reset)
11. Finite State Machine Design.
12. Design of FIFO
13. Design and interfacing of stepper motors
14. Design and implementation of Elevator
15. Design and implementation of bus protocols

COURSE OUTCOMES:

By the end of the course, the student will be able to

- 1: Design of the Combinational logic gates concepts for real time applications.
2. By Design and implementing Sequential circuits, exposure is created for the academic projects or real time application.
3. Problem solving skills in digital design area using HDL
4. Restate the various combinational and sequential circuits design and simulate using CAD tools .